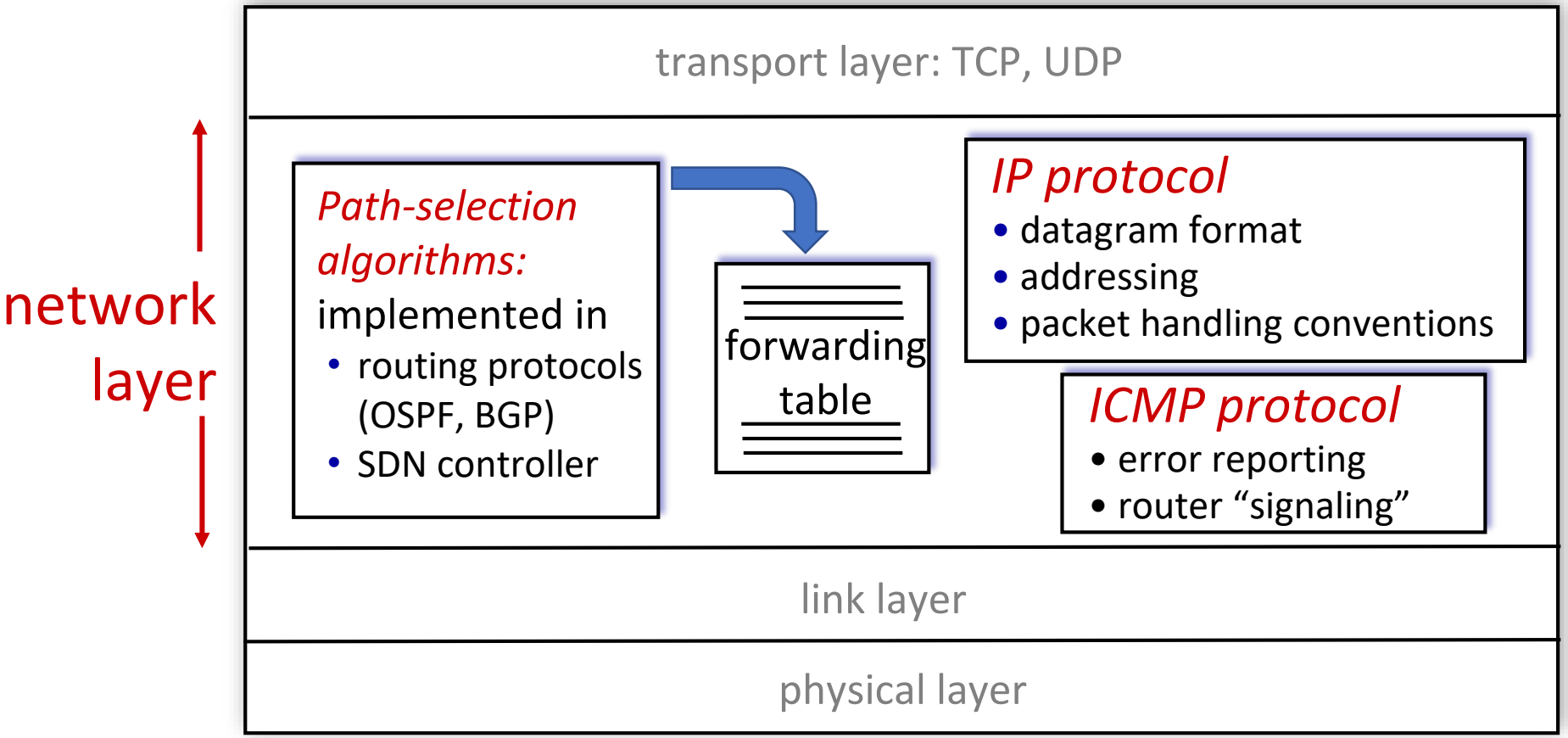
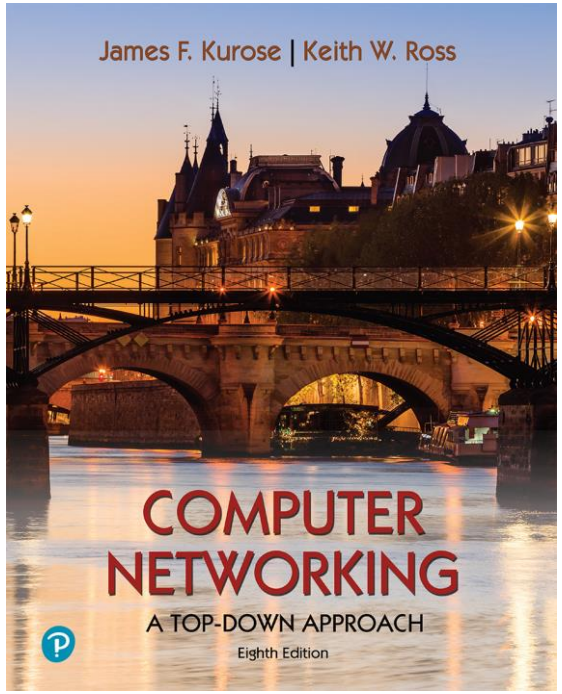
Chapter 4
Network Layer: Data Plane





Computer Networking: A Top-Down Approach
8th edition
Jim Kurose, Keith Ross
Pearson, 2020

Lecture 15

Network layer: “data plane” roadmap

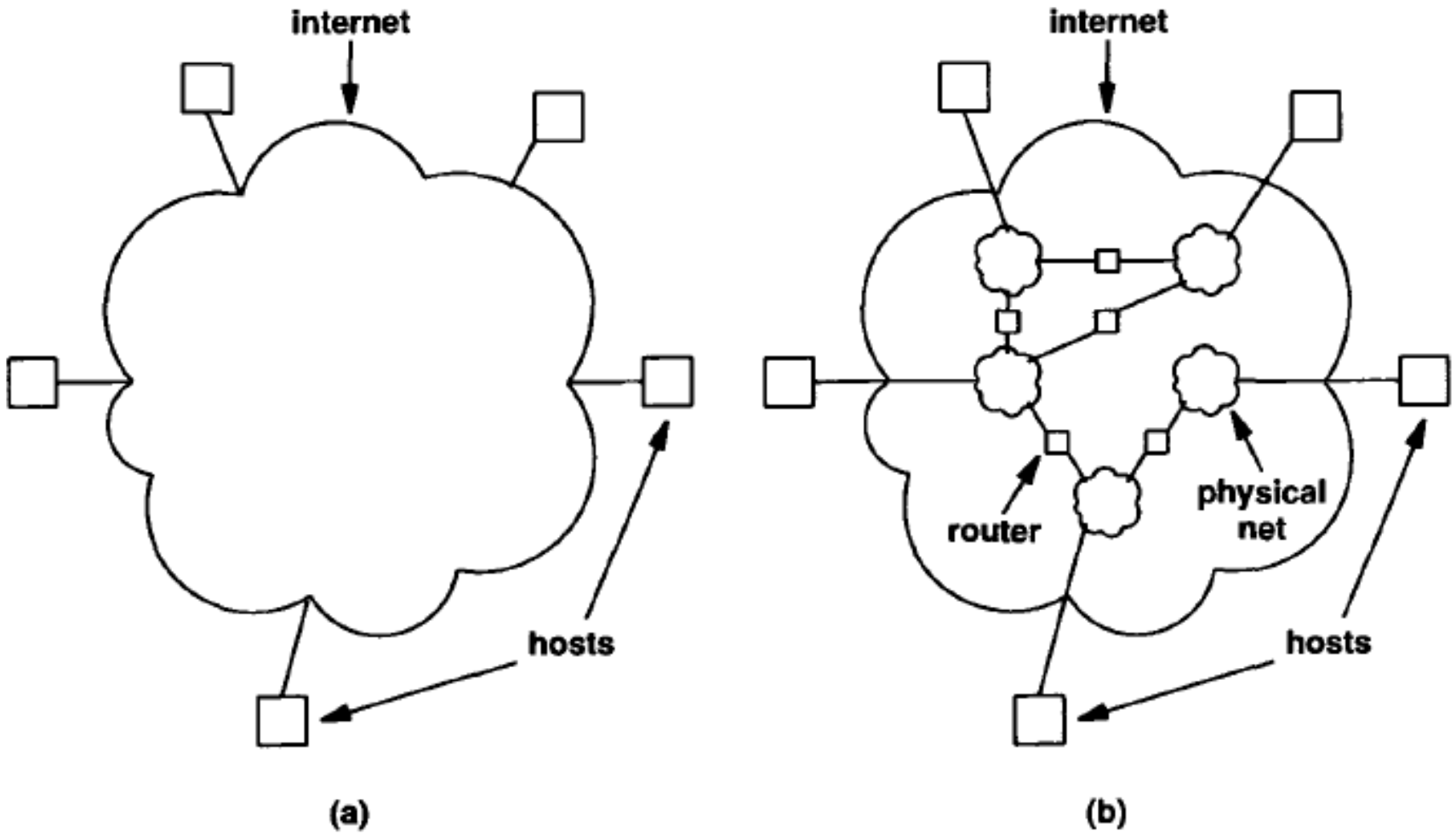
- Network layer: overview
  - data plane
  - control plane
- What’s inside a router
  - input ports, switching, output ports
  - buffer management, scheduling
- IP: the Internet Protocol
  - datagram format
  - addressing
  - network address translation
  - IPv6

- Generalized Forwarding, SDN
  - match+action
  - OpenFlow: match+action in action
- Middleboxes

Virtual Network

- Routers connect multiple physical networks.
  - Routers use the destination network, not the destination computer, when forwarding a packet.
- A user thinks of an internet as a single virtual network that interconnects all the hosts, and through which communication is possible; its underlying architecture is both hidden and irrelevant.
- We want a system that hides the details of underlying network hardware while providing universal communication services.

The User's View



(a) The user's view of a TCP/IP internet in which each computer appears to attach to a single large network, and (b) the structure of physical networks and routers that provide interconnection.

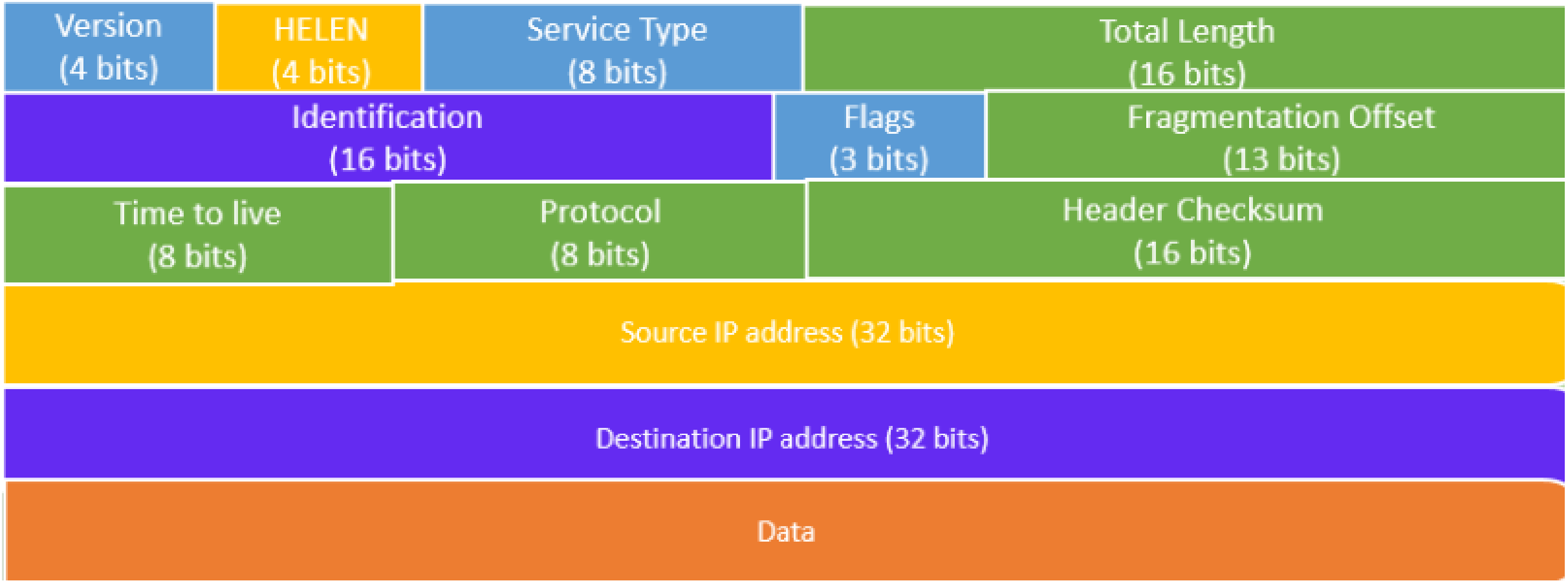
Network level interconnection

- The alternative to providing interconnection with application-level programs is a system based on network-level interconnection.
- A network-level interconnection provides a mechanism that delivers small packets of data from their original source to their ultimate destination without using intermediate application programs.

Advantages of using datagrams

- Switching small units of data instead of files or large messages has several advantages.
  - **First**, the scheme maps directly onto the underlying network hardware, making it extremely efficient.
  - **Second**, network-level interconnection separates data communication activities from application programs, permitting intermediate computers to handle network traffic without understanding the applications that are sending or receiving it.
  - **Third**, using network connections keeps the entire system flexible, making it possible to build general purpose communication facilities.
  - **Fourth**, the scheme allows network managers to add new network technologies by modifying or adding a single piece of new network level software, while application programs remain unchanged.

Network Layer: IP Header



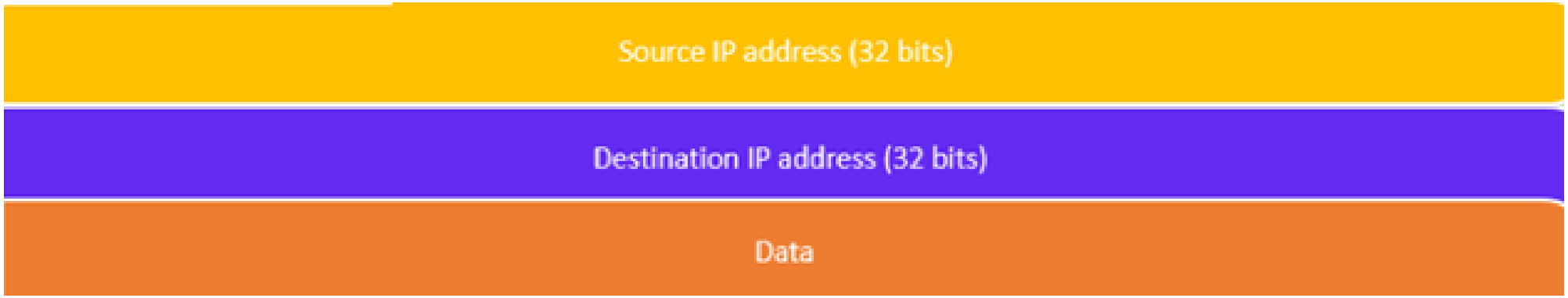
Version (4 bits)	HELEN (4 bits)	Service Type (8 bits)	Total Length (16 bits)

- **Version:** The first IP header field is a 4-bit version indicator. In IPv4, the value of its four bits is set to 0100, which indicates 4 in binary. However, if the router does not support the specified version, this packet will be dropped.
- **Internet Header Length:** Internet header length, shortly known as IHL, is 4 bits in size. It is also called HELEN (Header Length). This IP component is used to show how many 32-bit words are present in the header.
- **Type of Service:** Type of Service is also called Differentiated Services Code Point or DSCP. This field is provided features related to the quality of service for data streaming or VoIP calls. The first 3 bits are the priority bits. It is also used for specifying how you can handle Datagram.
- **Total length:** The total length is measured in bytes. The minimum size of an IP datagram is 20 bytes and the maximum, it can be 65535 bytes . HELEN and Total length can be used to calculate the dimension of the payload. All hosts are required to be able to read 576-byte datagrams. However, if a datagram is too large for the hosts in the network, the fragmentation method is widely used.

- **Identification:** Identification is a packet that is used to identify fragments of an IP datagram uniquely. Some have recommended using this field for other things like adding information for packet tracing, etc.
- **IP Flags:** Flag is a three-bit field that helps you to control and identify fragments. The following can be their possible configuration:
- Bit 0: is reserved and has to be set to zero
- Bit 1: means do not fragment (**DF**)
- Bit 2: means more fragments. (**MF**)
- **Fragment Offset:** Fragment Offset represents the number of Data Bytes ahead of the particular fragment in the specific Datagram. It is specified in terms of the number of 8 bytes, which has a maximum value of 65,528 bytes.

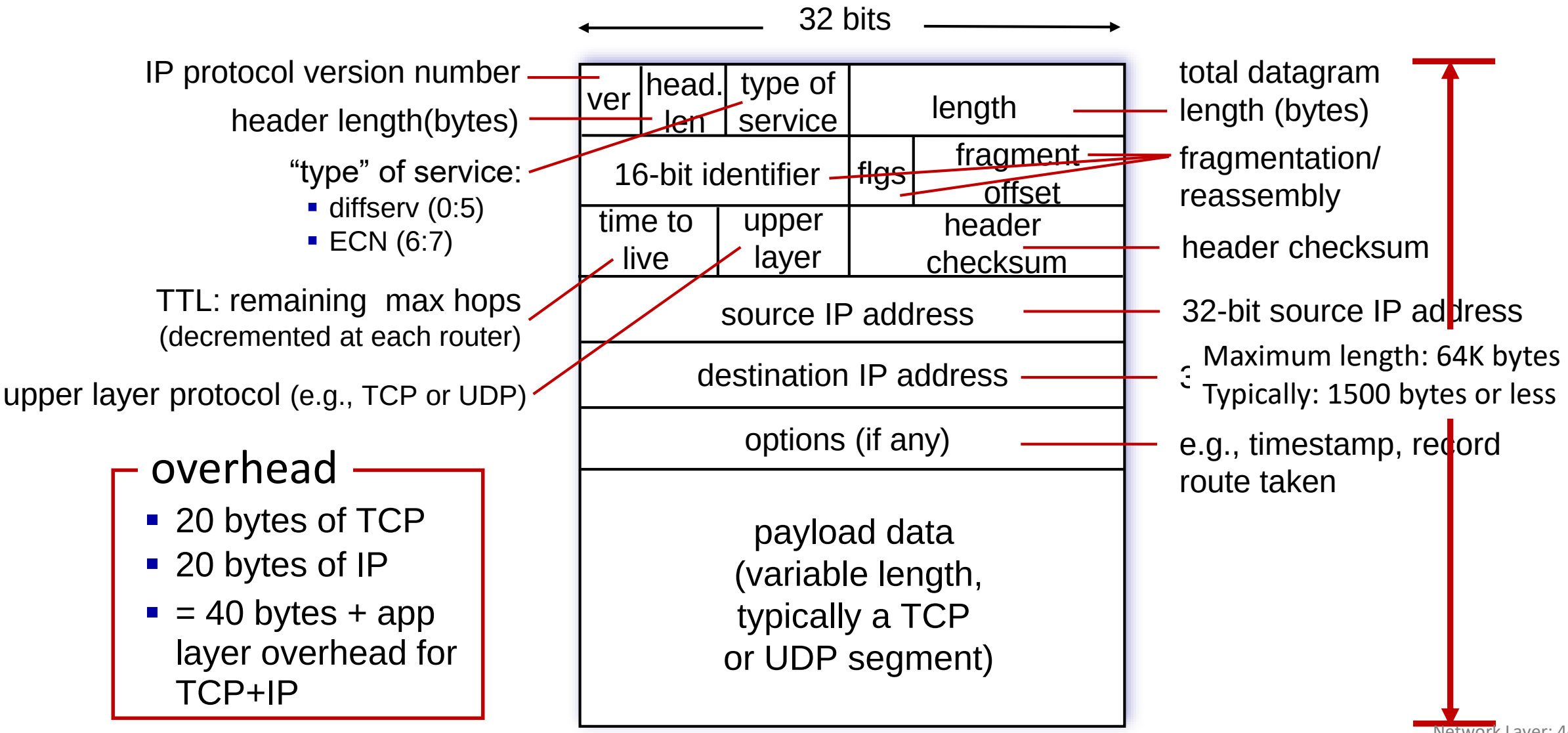
Time to live (8 bits)	Protocol (8 bits)	Header Checksum (16 bits)

- **Time to live:** It is an 8-bit field that indicates the maximum time the Datagram will be live in the internet system. The time duration is measured in seconds, and when the value of TTL is zero, the Datagram will be erased. Every time a datagram is processed its TTL value is decreased by one second. TTL are used so that datagrams are not delivered and discarded automatically. The value of TTL can be 0 to 255.
- **Protocol:** This IPv4 header is reserved to denote that internet protocol is used in the latter portion of the Datagram. For Example, 6 number digit is mostly used to indicate TCP, and 17 is used to denote the UDP protocol.
- **Header Checksum:** The next component is a 16 bits header checksum field, which is used to check the header for any errors. The IP header is compared to the value of its checksum. When the header checksum is not matching, then the packet will be discarded.



- **Source Address:** The source address is a 32-bit address of the source used for the IPv4 packet.
- **Destination address:** The destination address is also 32 bit in size stores the address of the receiver.
- **IP Options:** It is an optional field of IPv4 header used when the value of IHL (Internet Header Length) is set to greater than 5. It contains values and settings related with security, record route and time stamp, etc. You can see that list of options component ends with an End of Options or EOL in most cases.
- **Data:** This field stores the data from the protocol layer, which has handed over the data to the IP layer.

IP Datagram format summary

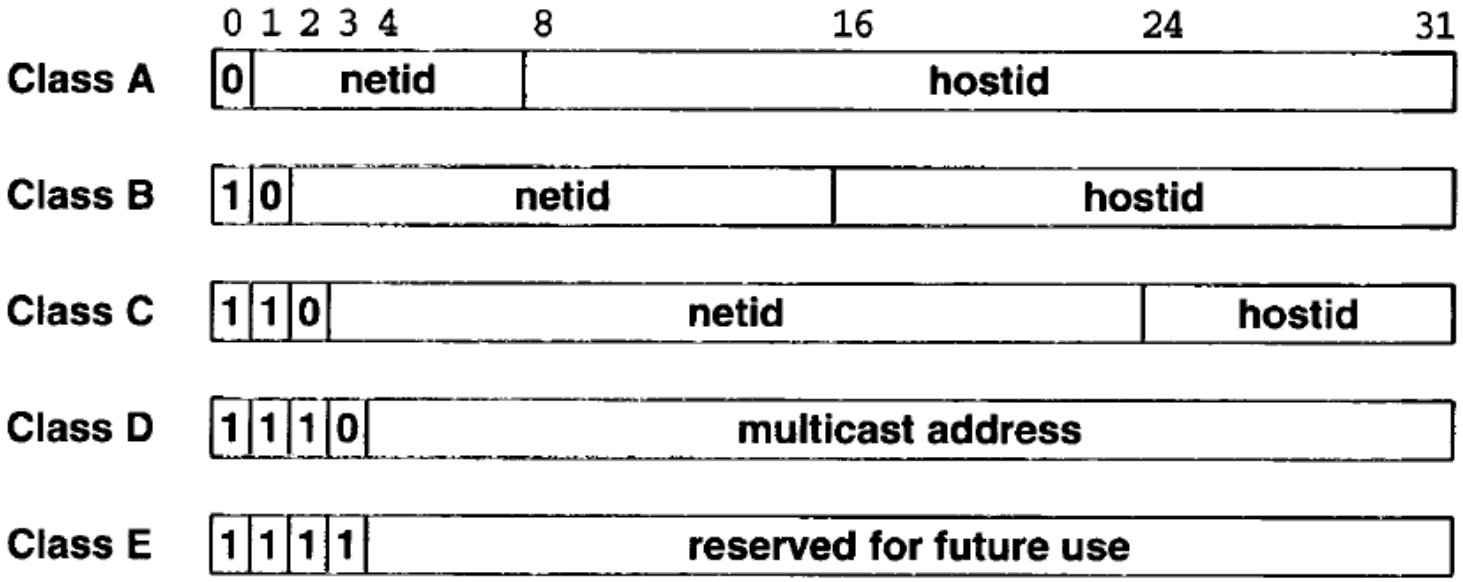


overhead

- 20 bytes of TCP
- 20 bytes of IP
- = 40 bytes + app layer overhead for TCP+IP

IP addressing: introduction

- IP address: 32-bit identifier associated with each host or router *interface*
- *Each host on a **TCP/IP** internet is assigned a unique 32-bit internet address that is used in all communication with that host.*



Class	Lowest Address	Highest Address
A	1.0.0.0	126.0.0.0
B	128.1.0.0	191.255.0.0
C	192.0.1.0	223.255.255.0
D	224.0.0.0	239.255.255.255
E	240.0.0.0	255.255.255.254

IP addressing: Subnet mask

- A subnet mask is a 32-bit number created by setting host bits to all 0s and setting network bits to all 1s. In this way, the subnet mask separates the IP address into the network and host addresses.
- The “255” address is always assigned to a broadcast address, and the “0” address is always assigned to a network address. Neither can be assigned to hosts, as they are reserved for these special purposes.
- Class A: 255.0.0.0
- Class B: 255.255.0.0
- Class C: 255.255.255.0

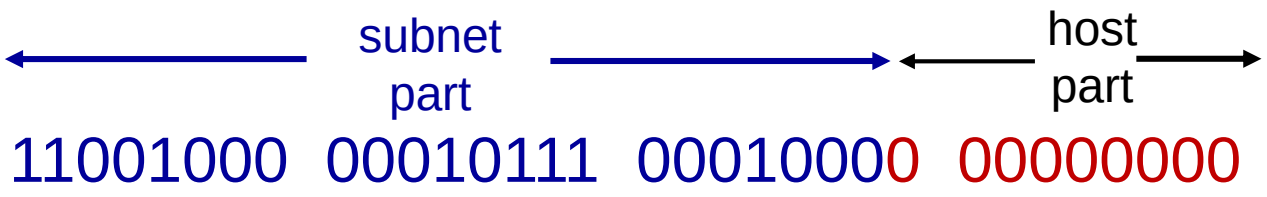
IP addressing: CIDR

- A system called **Classless Inter-Domain Routing** (CIDR), was developed as an alternative to traditional subnetting. You can add a specification in the IP address itself as to the number of significant bits that make up the routing or networking portion.
- For example, we could express the idea that the IP address 192.168.0.15 is associated with the netmask 255.255.255.0 by using the CIDR notation of 192.168.0.15/24. This means that the first 24 bits of the IP address given are considered significant for the network routing.

IP addressing: CIDR

CIDR: Classless InterDomain Routing (pronounced "cider")

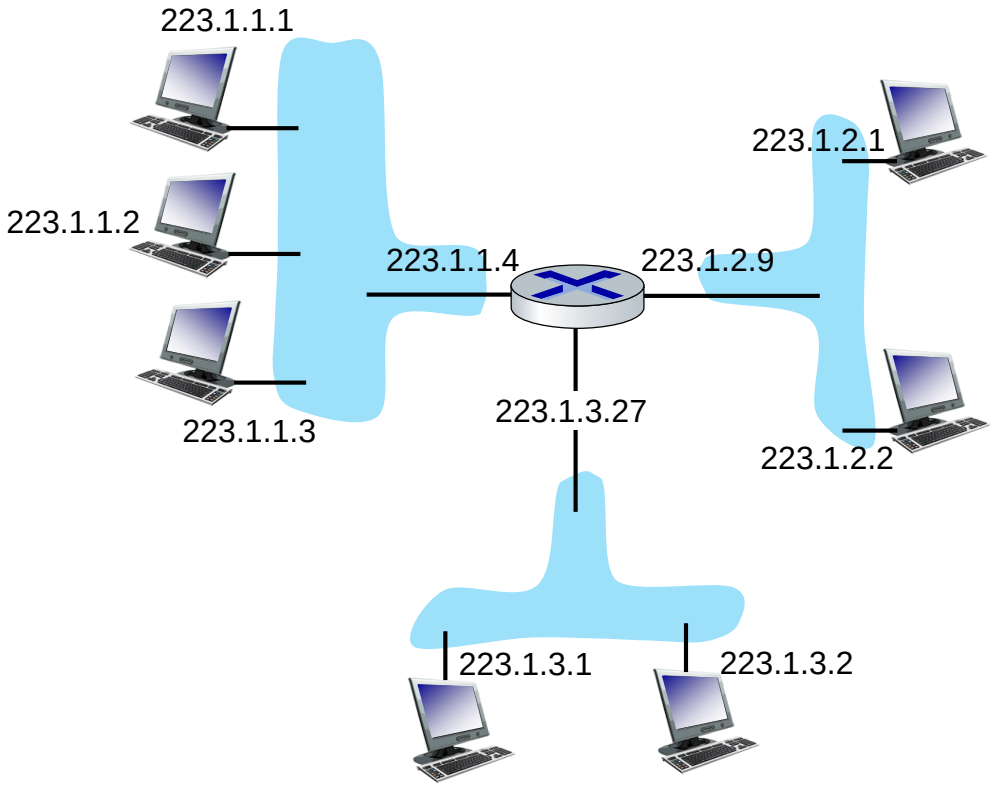
- subnet portion of address of arbitrary length
- address format: a.b.c.d/x, where x is # bits in subnet portion of address



200.23.16.0/23

IP addressing: introduction

- IP address: 32-bit identifier associated with each host or router *interface*
- interface: connection between host/router and physical link
  - router's typically have multiple interfaces
  - host typically has one or two interfaces (e.g., wired Ethernet, wireless 802.11)



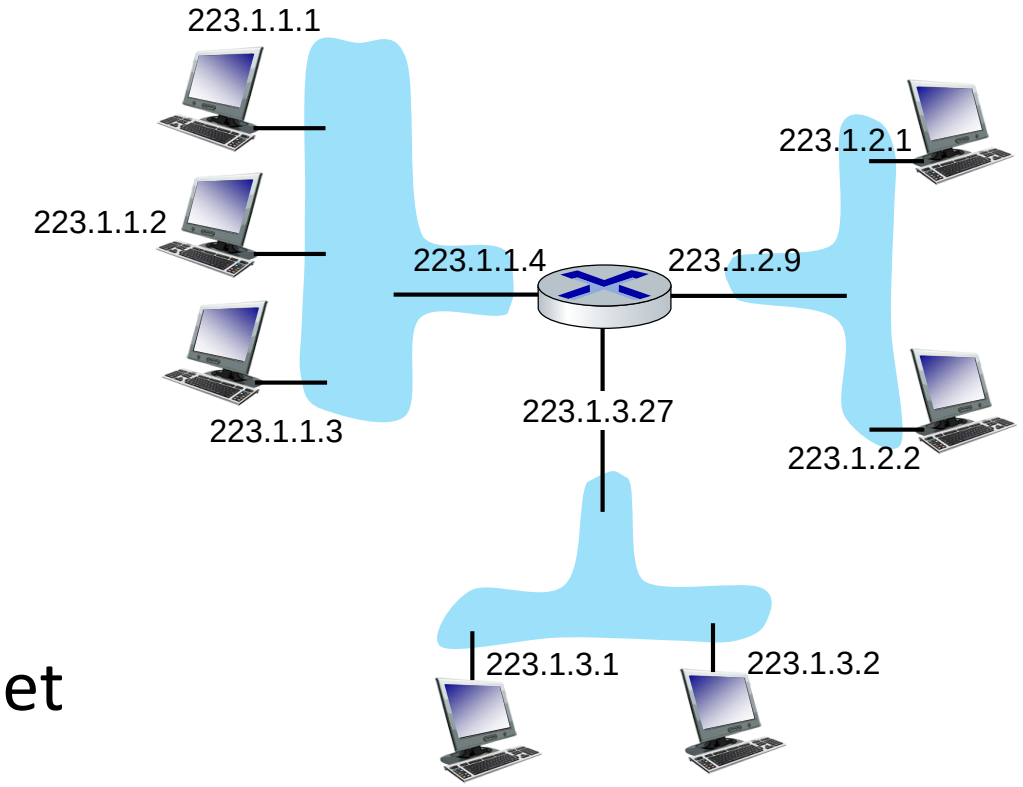
dotted-decimal IP address notation:

223.1.1.1 = 11011111 00000001 00000001 00000001

223 1 1 1

Subnets

- *What's a subnet ?*
  - device interfaces that can physically reach each other without passing through an intervening router
- IP addresses have structure:
  - subnet part: devices in same subnet have common high order bits
  - host part: remaining low order bits

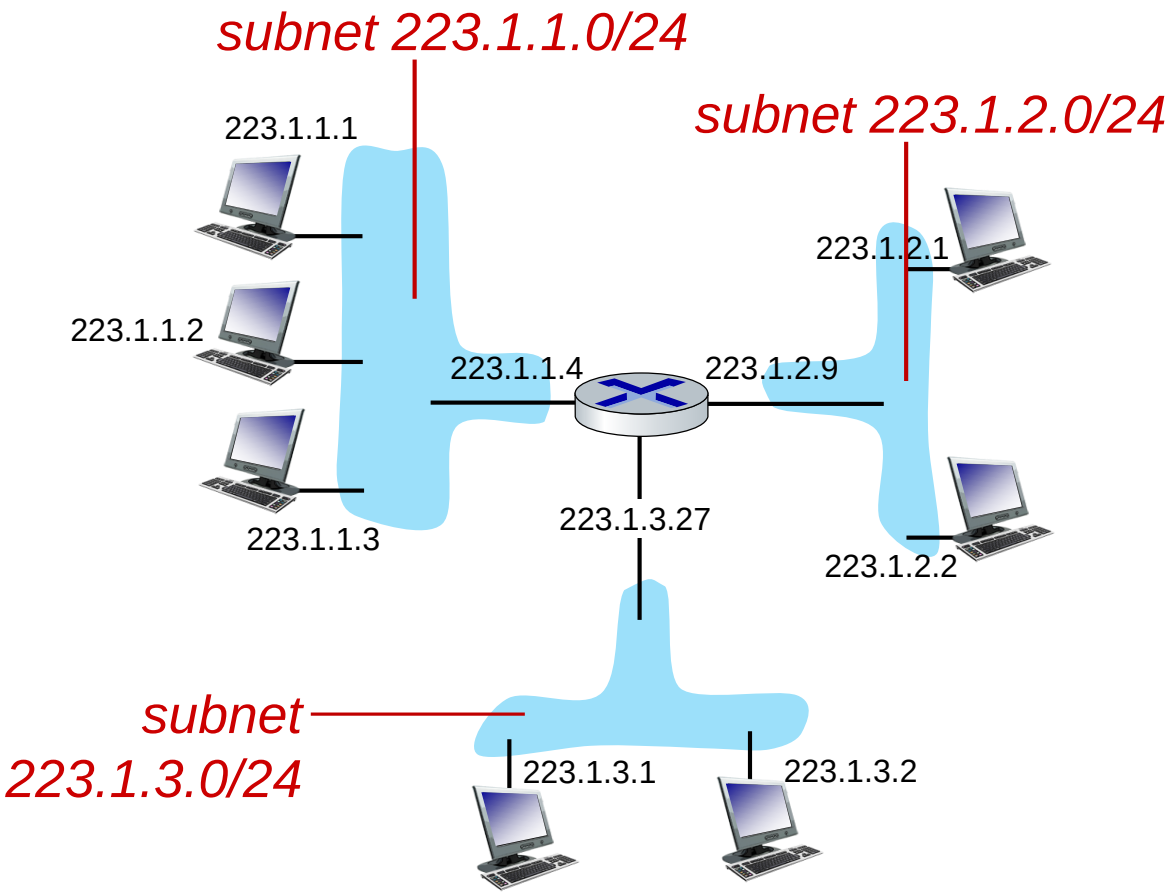


network consisting of 3 subnets

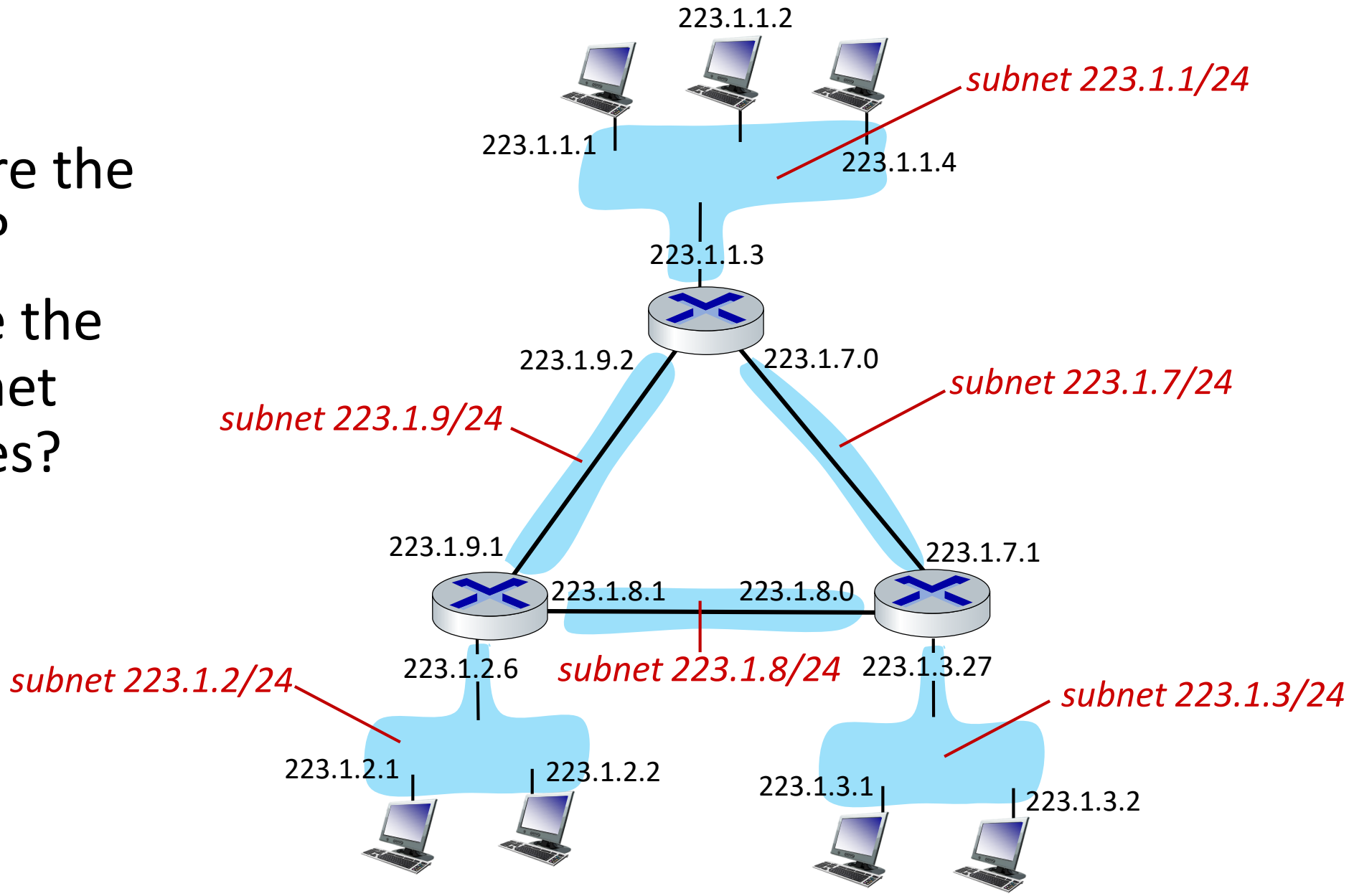
Subnets

Recipe for defining subnets:

- detach each interface from its host or router, creating "islands" of isolated networks
- each isolated network is called a *subnet*



subnet mask: /24
(high-order 24 bits: subnet part of IP address)

Subnets

- where are the subnets?
- what are the /24 subnet addresses?